Before you go in

15 minutes, in order. Don't learn anything new now: just remind yourself.

- Bag: printed pack + notes, squared paper, ruler, 2 pencils, eraser, pen, student ID.
- **Phone OFF and in the bag before you enter** — no electronic devices, no calculator.
- In the hall: read all tasks, start with the calculation you're surest of.
- Every task: **formula → steps table → drawing → boxed answer + convention**.

Numbers to know cold

4K @60 Hz	**$1.8 \cdot 10^9$ B/s (slide)**
1920×1080 @80 Hz	**6.03 ns / pixel**
100 circles + 100 triangles	**1200 + 2400 = 3600 B**
1024×768×24	**2.36 MB**
42″ 1920×1080	**52.45 ppi**
30 cm @ 50 cm	**2063 (slide 2069)**
3 bits/ch	**512 colours**
143, 240, 128 → 4 bit	**8, 15, 8**
[128,0,128] grey	**53 (avg 85)**
[255,255,0] grey	**226 (avg 170)**
bilinear 30/50/45/60 ¾,¾	**53.44**

Displays + image

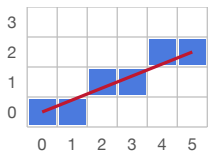
- rate = W·H·B·Hz
- t_px = 1/(W·H·Hz)
- memory = W·H·bits/8 → MB = $/10^6$
- vector: circle 3, triangle 6, rect 8 numbers × 4 B
- ppi: width = d·a/$\sqrt{(a^2+b^2)}$; 16:9 → d·0.8716
- eye: n = W / (D·0.00029)
- bilinear: P1 = αB+(1−α)A, P2 = αD+(1−α)C, N = βP2+(1−β)P1
- 2:1 down = average 2×2 block

Colour

- colours = $(2^b)^3$
- 8→n bits: floor(v / 2^(8−n))
- CMY = 255 − RGB
- grey = .299R + .587G + .114B (and (R+G+B)/3)
- mix = $\alpha C_1 + (1-\alpha)C_2$
- HSV: V = max, S = (max−min)/max, H: R 0°, Y 60°, G 120°, C 180°, B 240°, M 300°
- cone: tip black, axis greys, top rim = pure colours
- CIE: mix of 2 = straight segment; 3 = triangle (gamut)
- cube: black origin, white opposite, yellow (1,1,0)

DDA + Bresenham

- DDA: m = Δy/Δx, x+1, y += m, round; only |m| ≤ 1 (else gaps)
- **Bresenham**: start at the smaller x
- a = 2Δy, b = 2Δy − 2Δx, p_0 = 2Δy − Δx
- p < 0 → (x+1, y), p += a
- p ≥ 0 → (x+1, y+1), p += b (**0 goes up**)
- (0,0)→(5,2): p −1, 3, −3, 1, −5 → (0,0)(1,0)(2,1)(3,1)(4,2)(5,2)
- Antialias: 2 nearest pixels per column, intensity = 1 − distance, sum 100 %



Circle + fills

- Circle: (0,r), $p_0 = 5/4 - r$
- $p < 0 \rightarrow (x+1, y)$, $p \mathrel{+}= 2x_+ + 1$
- $p \geq 0 \rightarrow (x+1, \mathbf{y-1})$, $p \mathrel{+}= 2x_+ + 1 - 2y_+$ (NEW x, y)
- stop $x \geq y$, mirror 8 ways
- r = 10: (0,10)(1,10)(2,10)(3,10)(4,9)(5,9)(6,8)(7,7)
- Scan-line: crossings → sort → fill pairs; peak counts twice
- Seed fill: recursive L, R, U, D; name the seed

Barycentric

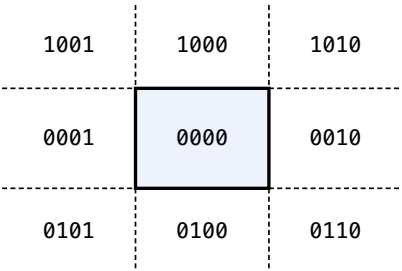
- $U = \alpha A + \beta B + \gamma C$, $\alpha + \beta + \gamma = 1$
- inside $\Leftrightarrow$ all ≥ 0 (0 = on an edge)
- value = $\alpha f_A + \beta f_B + \gamma f_C$
- A(0,0) B(0,4) C(4,0), U(1,2): $\gamma = 1/4$, $\beta = 2/4$, $\alpha = 1/4 \rightarrow$ inside
- grey 255/0/128 $\rightarrow$ 96 · 10/25/128 $\rightarrow$ 47
- texture: same weights on (u,v)

Transforms + frame

- Column vectors: **M = last · … · first**
- T = [1 0 dx; 0 1 dy; 0 0 1]
- S = [sx 0 0; 0 sy 0; 0 0 1]
- R = [cos −sin 0; sin cos 0; 0 0 1] (CCW)
- about point p: T(p) · X · T(−p)
- cos 45° = sin 45° = 0.7071
- Frame: rows = new axes; frame +θ = point −θ
- P(2,2), frame +90° → **(2, −2)**

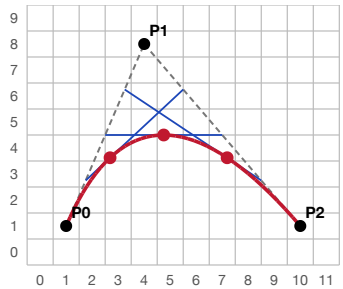
Clipping

- Code **T B R L**: 1001 1000 1010 / 0001 0000 0010 / 0101 0100 0110
- both 0000 → accept; AND ≠ 0 → reject
- else cut an outside end: left, top, right, bottom; recode; repeat
- x = c: $y = y_1 + (y_2 - y_1)(c - x_1)/(x_2 - x_1)$
- S–H: one edge at a time; out→in: P, W · in→in: W · in→out: P · out→out: —
- (7,1)(1,4)(5,10) → (4,2.5)(4,5)(6.11,5)(7,1)



Bézier

- degree = points − 1; through P_0, P_n only
- quadratic: $(1-t)^2P_0 + 2t(1-t)P_1 + t^2P_2$
- cubic: $(1-t)^3$, $3t(1-t)^2$, $3t^2(1-t)$, t^3
- t = ½ quadratic: $(P_0 + 2P_1 + P_2)/4$
- t = ½ cubic: $(P_0 + 3P_1 + 3P_2 + P_3)/8$
- de Casteljau: split each segment at t, repeat
- (1,1)(4,8)(10,1): t = .5 → (4.75, 4.5)
- C1 join: $Q_1 = 2P_3 - P_2$



Top traps

1. bits vs bytes (÷8)
2. MB 10^6 vs MiB 2^{20} — give both
3. $(2^b)^3$, not 2^b
4. Bresenham: p = 0 goes UP
5. circle: y−1, NEW x, y
6. AND = 0000 ≠ accept
7. first transform = rightmost matrix
8. sin sign in R
9. frame = point rotated −θ
0. pixel (1,2), not (1,1)
1. ns, not μs
2. 55/9 = 6.11, not 6

You are ready

You have practised every task type with new numbers. In the test:

1. Breathe. Read all the tasks.
2. Sure points first, then the rest.
3. Stuck? Write the formula and the method anyway: partial credit.
4. Last 5 min: check units, signs, coordinates, and box the answers.

Powodzenia! 🍀